

EECE 306 Lab 4 notebook. The Quadrature Engine and Continuous Charge Distributions

Team 8. Thalea Collymore, Kroix Jones, Xavier Moore

Fill in every TODO, run `publish('Lab04_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

Contents

- [Setup](#)
- [Geometry before physics](#)
- [Observed order of each rule](#)
- [The Gauss rule does not fit the same mold](#)
- [Finite line against the infinite line formula](#)
- [Disk against the infinite sheet formula](#)
- [A nonuniform density through the same constructor](#)
- [Interpretation](#)
- [Problems encountered](#)
- [Full test suite](#)

Setup

```
clear; close all;
e0 = em.const.eps0();
```

Geometry before physics

The sum of the weights must return length, area, and volume with no field anywhere in sight.

```
circ = @(t)[cos(t) sin(t) zeros(size(t))];
[~, wc] = em.quad.line(circ, [0 2*pi], 400);
sph = @(th,ph)[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
[~, ws] = em.quad.surf(sph, [0 pi], [0 2*pi], 60, 120);
ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
[~, wv] = em.quad.vol(ball, [0 pi], [0 2*pi], [0 1], 30, 60, 30);
sq = @(u,v)[u v 0];
[~, wq] = em.quad.surf(sq, [0 1], [0 1], 20, 20);
fprintf('circle length %.6f expect %.6f\n', sum(wc), 2*pi);
fprintf('sphere area %.6f expect %.6f\n', sum(ws), 4*pi);
fprintf('ball volume %.6f expect %.6f\n', sum(wv), 4*pi/3);
fprintf('square area %.6f expect %.6f\n', sum(wq), 1);
```

```
circle length 6.283185 expect 6.283185
sphere area 12.567806 expect 12.566371
ball volume 4.189541 expect 4.188790
square area 1.000000 expect 1.000000
```

Observed order of each rule

Errors are collected in a plain loop, then the standard fitting tool reports the observed order. One representative log log plot is shown.

```
Iex = exp(1) - 1;
Ns = [8 16 32 64 128]';
rules = {'midpoint', 'trapz', 'simpson'};
for k = 1:3
    rule = rules{k};
    errs = zeros(size(Ns));
    for j = 1:numel(Ns)
        [t, w] = em.quad.nodes(0, 1, Ns(j), rule);
        errs(j) = abs(sum(exp(t) .* w) - Iex);
    end
```

```

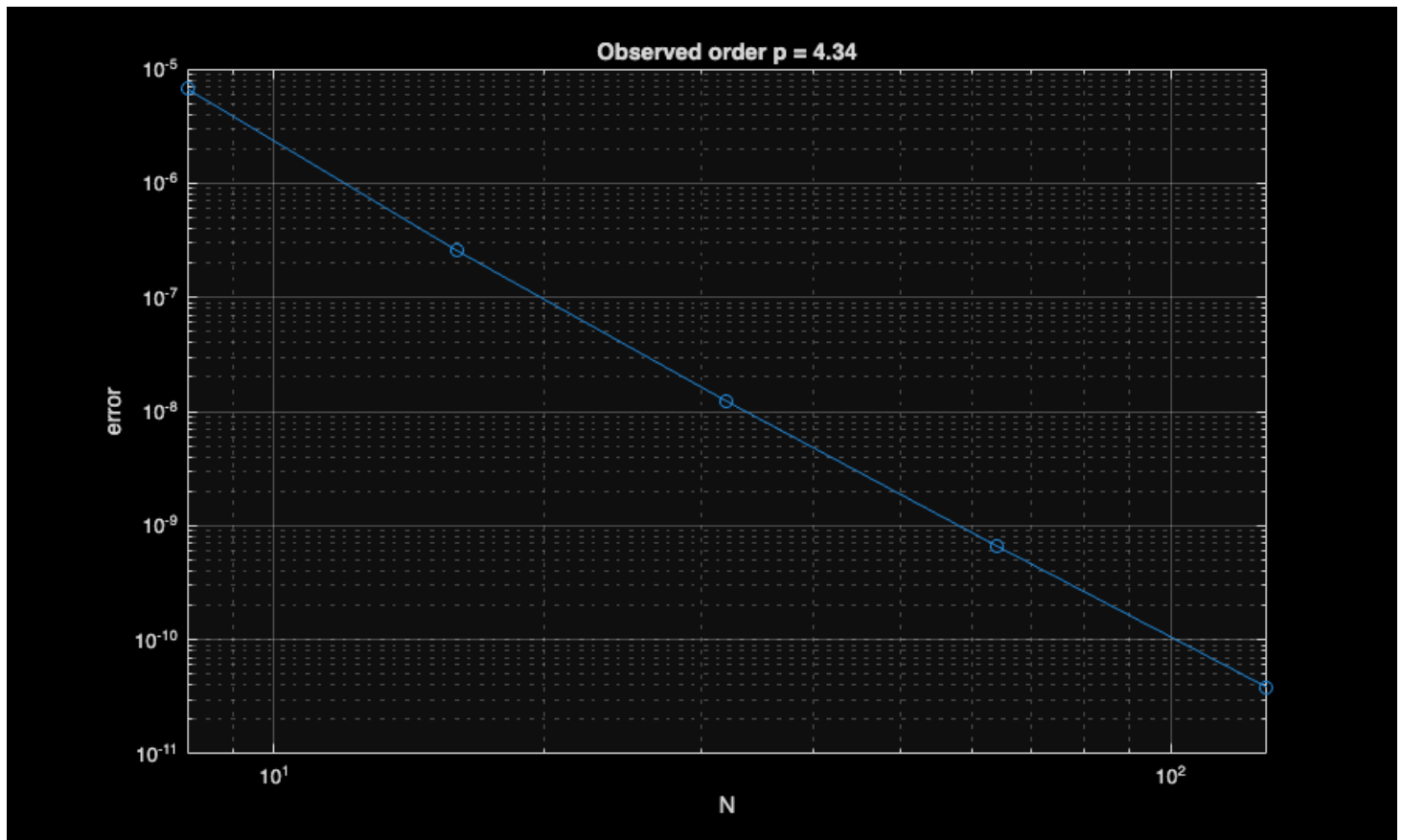
p = em.test.convergence(@(N) errs(Ns == N), Ns, struct('plot', strcmp(rule, 'simpson')));
fprintf('%-9s observed order p = %.2f\n', rule, p);
end

```

```

midpoint observed order p = 2.00
trapz     observed order p = 2.09
simpson   observed order p = 4.34

```



The Gauss rule does not fit the same mold

```

for N = [2 4 8 16]
    [t, w] = em.quad.nodes(0, 1, N, 'gauss');
    fprintf('gauss N = %2d   error = %.3e\n', N, abs(sum(exp(t) .* w) - Iex));
end

```

% An N-point Gauss rule is exact for every polynomial through degree
 % $2N - 1$, so a smooth nonpolynomial integrand can converge faster than
 % any single fixed power of N. Its errors therefore do not follow one
 % straight line on a log-log plot and eventually reach floating-point
 % precision, so a fitted slope is not a meaningful fixed order.

```

gauss N = 2   error = 3.855e-04
gauss N = 4   error = 9.330e-10
gauss N = 8   error = 1.110e-15
gauss N = 16  error = 3.553e-15

```

Finite line against the infinite line formula

```

d = 1;
Lod = logspace(0, 2, 15)';
ratio = zeros(size(Lod));
for k = 1:numel(Lod)
    L = Lod(k) * d;

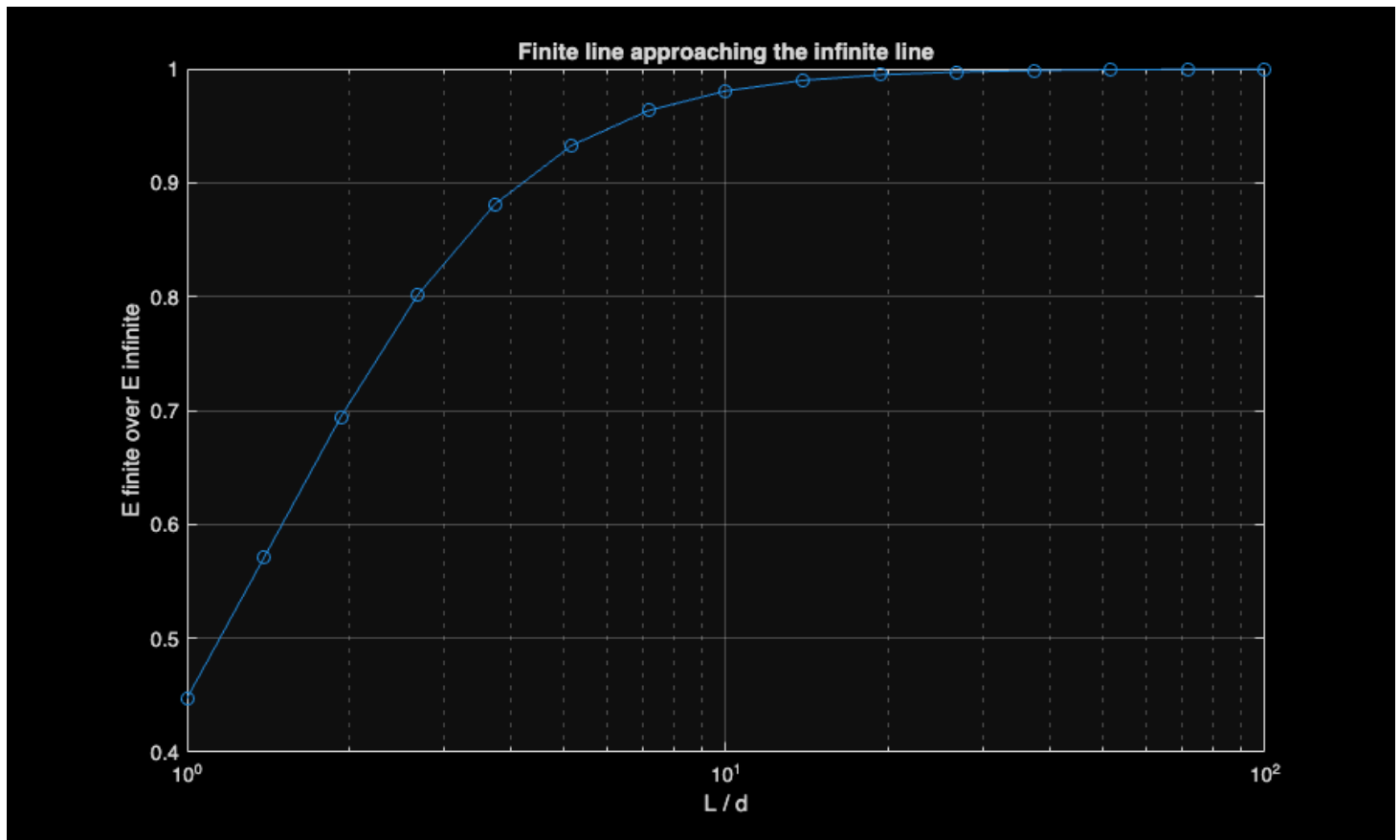
```

```

lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
sl = em.src.lineCharge(1e-9, lineC, [-L/2 L/2], 800);
Ef = em.field.E(sl, [d 0 0]);
ratio(k) = Ef(1) / (1e-9 / (2*pi*e0*d));
end
figure; semilogx(Lod, ratio, 'o-'); grid on
xlabel('L / d'); ylabel('E finite over E infinite')
title('Finite line approaching the infinite line')
iAbove = find(ratio > 0.99, 1);
fprintf('ratio first exceeds 0.99 at L/d about %.1f\n', Lod(iAbove));

```

ratio first exceeds 0.99 at L/d about 19.3



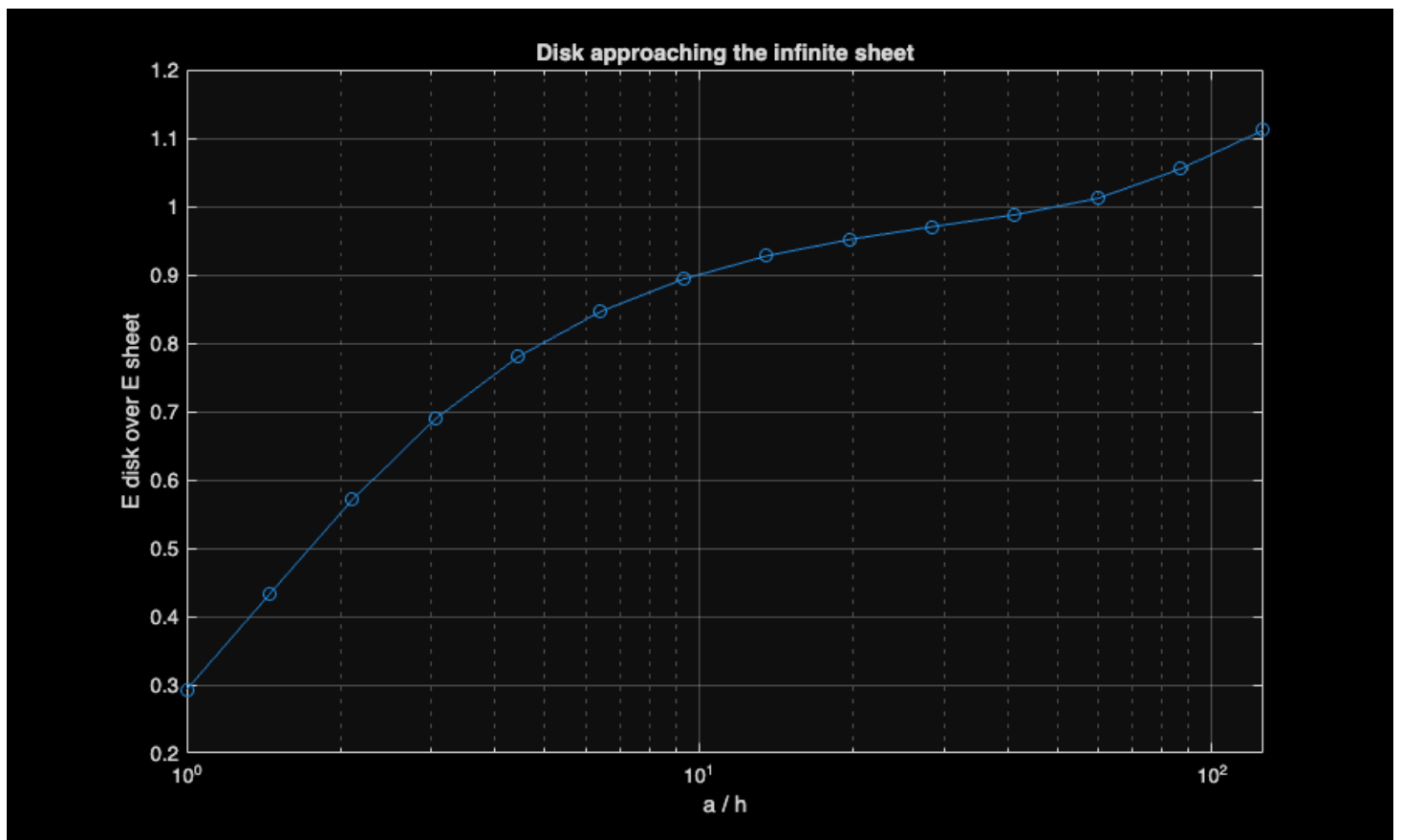
Disk against the infinite sheet formula

```

h = 1;
aoh = logspace(0, 2.1, 14)';
ratioD = zeros(size(aoh));
for k = 1:numel(aoh)
    a = aoh(k) * h;
    disk = @(u,v)[u.*cos(v) u.*sin(v) zeros(size(u))];
    sd = em.src.surfCharge(1e-9, disk, [0 a], [0 2*pi], 80, 80);
    Ed = em.field.E(sd, [0 0 h]);
    ratioD(k) = Ed(3) / (1e-9 / (2*e0));
end
figure; semilogx(aoh, ratioD, 'o-'); grid on
xlabel('a / h'); ylabel('E disk over E sheet')
title('Disk approaching the infinite sheet')
iA = find(ratioD > 0.99, 1);
fprintf('ratio first exceeds 0.99 at a/h about %.1f\n', aoh(iA));

```

ratio first exceeds 0.99 at a/h about 59.8



A nonuniform density through the same constructor

```
lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
sn = em.src.lineCharge(@(r) r(:,3), lineC, [-1 1], 400);
fprintf('net charge of rho(z) = z on [-1, 1] = %.3e C (expect 0)\n', sum(sn.q .* sn.w));
```

```
net charge of rho(z) = z on [-1, 1] = -2.602e-17 C (expect 0)
```

Interpretation

The measured midpoint and trapezoid orders are near two, while Simpson's order is near four, which verifies the node and weight construction. On the sampled grids, the line first exceeds 0.99 at L/d about 19.3, while the disk first exceeds 0.99 at a/h about 59.8, so the disk approaches its infinite limit much more slowly. The supplied closed form places the exact disk threshold near $a/h = 100$, so the earlier numerical crossing also shows that the fixed 80-point radial grid becomes under-resolved as the disk grows. In the Lab 7 and Lab 8 numerical solvers, an unexpectedly low measured order would be an early sign of an incorrectly handled boundary or an off-by-one error in node placement.

Problems encountered

The starter study uses even values of N for Simpson's rule even though the composite 1/3 rule normally requires an odd number of nodes. We kept exactly N nodes by applying Simpson's 3/8 rule on the final three subintervals for even N , then verified that the combined rule retains fourth-order convergence.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
```

```
eta =
```

```
376.7303
```

```
test_lab01 checks complete
```

```
PASS test_lab01          0.06 s
test_lab02 checks complete
PASS test_lab02          0.03 s
5000-point field evaluation: 0.0029 s
test_lab03 checks complete
PASS test_lab03          0.03 s
test_lab04 checks complete
PASS test_lab04          0.18 s
```

```
4 passed, 0 failed, 4 total
All tests passing.
```